The Effects of Recovery and Workload on Player Physical Performance at the FIFA World Cup

2026-09-28

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggplot2)
library(psych)
```

```
##
## Attaching package: 'psych'
```

```
## The following objects are masked from 'package:ggplot2':
##
##     %+%, alpha
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
```

```
## The following objects are masked from 'package:psych':
##
##     alpha, rescale
```

```
library(splines)
library(lme4)
```

```
## Loading required package: Matrix
```

1. Data Processing

```
# Load raw data
df <- read.csv("raw_data.csv")
nrow(df)
```

```
## [1] 5287
```

```
# Variables whose previous-match values we want to retain
prev_vars <- c(
  "nominal_minutes_played",
```

```
  "total_distance_m",
  "zone1_distance_m",
  "zone2_distance_m",
  "zone3_distance_m",
  "zone4_distance_m",
  "zone5_distance_m",
  "top_speed_kmh"
)

# Add previous-match information
df <- df %>%
  mutate(
    # Convert match_date to Date
    match_date = as.Date(match_date)
  ) %>%

  # Put each player's appearances in chronological order
  arrange(
    tournament_year,
    player_tournament_id,
    match_date,
    fifa_match_number
  ) %>%

  # Previous matches must be computed separately within each tournament/player
  group_by(tournament_year, player_tournament_id) %>%

  mutate(
    # Match metadata from previous appearance
    prev_match_id = lag(match_id),
    prev_match_date = lag(match_date),

    # Days since previous appearance
    recovery_days = as.integer(match_date - prev_match_date),

    # Previous-match physical statistics
    across(
      all_of(prev_vars),
      lag,
      .names = "prev_{.col}"
    )
  ) %>%

  ungroup()
```

```
df <- df %>%
  # Filter out rows where player played less than 10 minutes
  filter(
    !(is.na(nominal_minutes_played) | nominal_minutes_played < 10)
  ) %>%

  # Normalize distance statistics by minutes played
  mutate(
    normalized_total_distance_m = total_distance_m / nominal_minutes_played,
```

```
    normalized_zone1_distance_m = zone1_distance_m / nominal_minutes_played,
    normalized_zone2_distance_m = zone2_distance_m / nominal_minutes_played,
    normalized_zone3_distance_m = zone3_distance_m / nominal_minutes_played,
    normalized_zone4_distance_m = zone4_distance_m / nominal_minutes_played,
    normalized_zone5_distance_m = zone5_distance_m / nominal_minutes_played,
  )

nrow(df)
```

```
## [1] 4876
```

```
head(df)
```

```
## # A tibble: 6 x 55
##   tournament_year match_id  fifa_match_number stage  stage_type group match_date
##             <int> <chr>                 <int> <chr>  <chr>      <chr> <date>
## 1            2022 2022_M008                 8 Group~ group      "C"   2022-11-22
## 2            2022 2022_M024                24 Group~ group      "C"   2022-11-26
## 3            2022 2022_M039                39 Group~ group      "C"   2022-11-30
## 4            2022 2022_M050                50 Round~ knockout   ""    2022-12-03
## 5            2022 2022_M057                57 Quart~ knockout   ""    2022-12-09
## 6            2022 2022_M064                64 Final  knockout   ""    2022-12-18
## # i 48 more variables: kickoff_time_local <chr>, stadium <chr>, team <chr>,
## #   opponent <chr>, team_goals <int>, opponent_goals <int>,
## #   team_tournament_id <chr>, player_name <chr>, player_tournament_id <chr>,
## #   shirt_number <int>, position <chr>, starter <int>, sub_on_minute <int>,
## #   sub_off_minute <int>, match_duration_min <int>,
## #   nominal_minutes_played <int>, total_distance_m <dbl>,
## #   zone1_distance_m <dbl>, zone2_distance_m <dbl>, zone3_distance_m <dbl>, ...
```

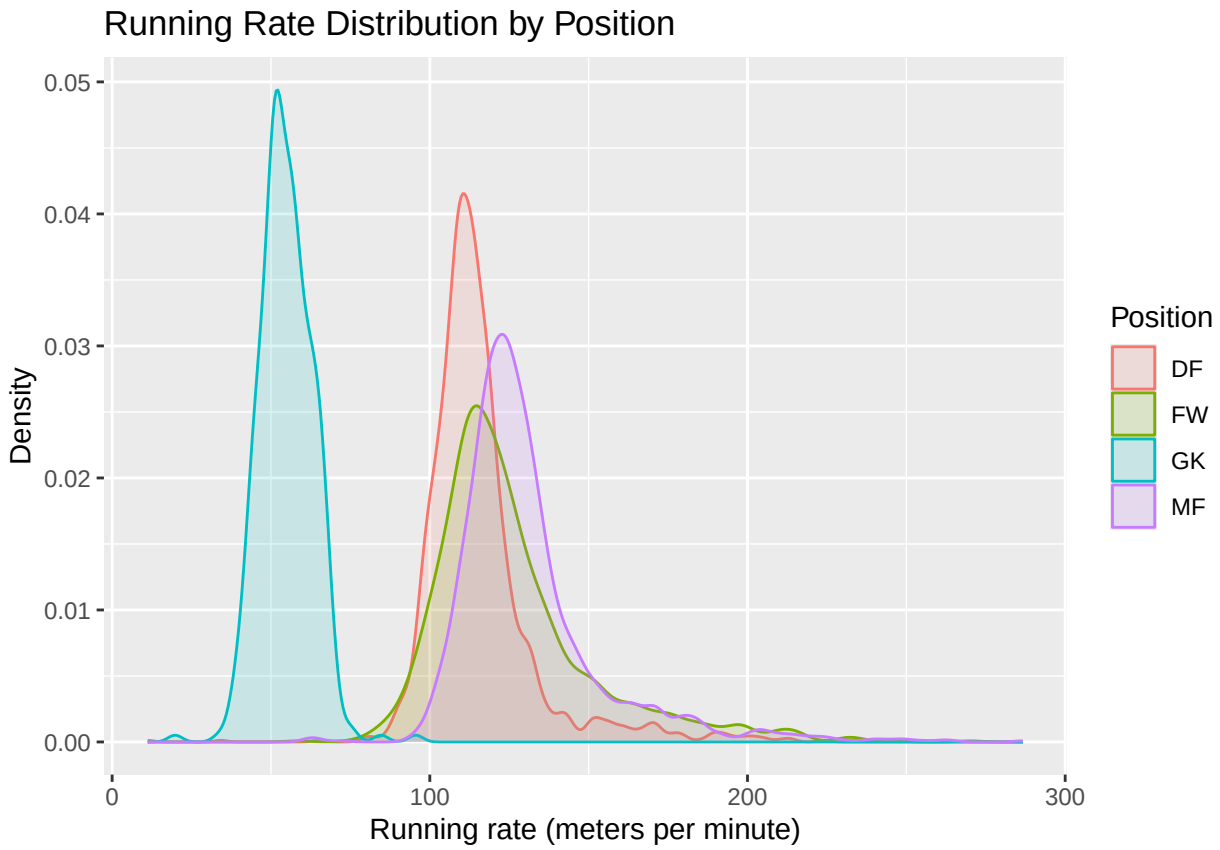
```
str(df)
```

```
## tibble [4,876 x 55] (S3: tbl_df/tbl/data.frame)
##  $ tournament_year            : int [1:4876] 2022 2022 2022 2022 2022 2022 2022 2022 2022 2022 ...
##  $ match_id                   : chr [1:4876] "2022_M008" "2022_M024" "2022_M039" "2022_M050" ...
##  $ fifa_match_number          : int [1:4876] 8 24 39 50 57 64 8 24 39 50 ...
##  $ stage                      : chr [1:4876] "Group Stage" "Group Stage" "Group Stage" "Round of 16"
##  $ stage_type                 : chr [1:4876] "group" "group" "group" "knockout" ...
##  $ group                      : chr [1:4876] "C" "C" "C" "" ...
##  $ match_date                 : Date[1:4876], format: "2022-11-22" "2022-11-26" ...
##  $ kickoff_time_local         : chr [1:4876] "13:00" "22:00" "22:00" "22:00" ...
##  $ stadium                    : chr [1:4876] "Lusail Stadium" "Lusail Stadium" "Stadium 974" "Ahmad
##  $ team                       : chr [1:4876] "Argentina" "Argentina" "Argentina" "Argentina" ...
##  $ opponent                   : chr [1:4876] "Saudi Arabia" "Mexico" "Poland" "Australia" ...
##  $ team_goals                 : int [1:4876] 1 2 2 2 2 3 1 2 2 2 ...
##  $ opponent_goals             : int [1:4876] 2 0 0 1 2 3 2 0 0 1 ...
##  $ team_tournament_id         : chr [1:4876] "2022_ARGENTINA" "2022_ARGENTINA" "2022_ARGENTINA" "202
##  $ player_name                : chr [1:4876] "ACUNA Marcos" "ACUNA Marcos" "ACUNA Marcos" "ACUNA Mar
##  $ player_tournament_id       : chr [1:4876] "2022_ARGENTINA_ACUNA_MARCOS_8" "2022_ARGENTINA_ACUNA_M
##  $ shirt_number               : int [1:4876] 8 8 8 8 8 8 9 9 9 9 ...
##  $ position                   : chr [1:4876] "MF" "MF" "MF" "MF" ...
##  $ starter                    : int [1:4876] 0 1 1 1 1 0 0 0 1 1 ...
##  $ sub_on_minute              : int [1:4876] 71 NA NA NA NA 64 59 63 NA NA ...
##  $ sub_off_minute             : int [1:4876] NA NA 59 72 78 NA NA NA 79 72 ...
##  $ match_duration_min         : int [1:4876] 90 90 90 90 120 120 90 90 90 90 ...
```

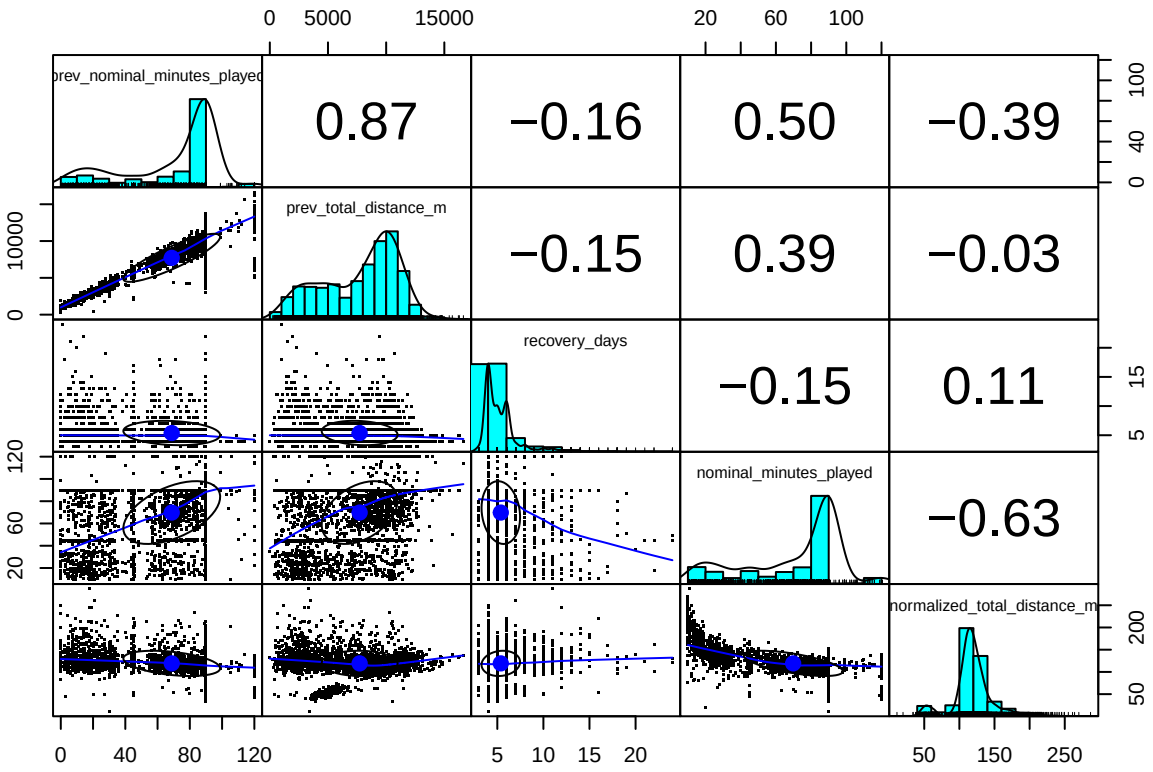
```
##  $ nominal_minutes_played     : int [1:4876] 19 90 59 72 78 56 31 27 79 72 ...
##  $ total_distance_m           : num [1:4876] 3237 9991 6790 7821 8271 ...
##  $ zone1_distance_m           : num [1:4876] 1220 3888 2330 2693 3164 ...
##  $ zone2_distance_m           : num [1:4876] 1302 4326 3036 3625 3417 ...
##  $ zone3_distance_m           : num [1:4876] 476 974 847 881 1015 ...
##  $ zone4_distance_m           : num [1:4876] 171 552 452 436 512 ...
##  $ zone5_distance_m           : num [1:4876] 69 251 126 186 163 ...
##  $ high_speed_runs            : int [1:4876] 39 101 73 88 86 95 51 51 114 103 ...
##  $ sprints                    : int [1:4876] 16 47 29 34 38 36 29 20 52 50 ...
##  $ top_speed_kmh              : num [1:4876] 31.9 29.6 30.7 29.5 31.6 30 31.6 30 32.3 32.4 ...
##  $ high_speed_runs_definition : chr [1:4876] "Not specified on table" "Not specified on table" "Zone
##  $ sprints_definition         : chr [1:4876] "Not specified on table" "Not specified on table" "Zone
##  $ source_pdf                 : chr [1:4876] "M08 ARG v KSA post-match report 1.1.pdf" "report_13301
##  $ lineup_page                : int [1:4876] 2 2 2 2 2 2 2 2 2 2 ...
##  $ physical_page              : int [1:4876] 50 50 52 50 51 51 50 50 52 50 ...
##  $ qc_flag                    : chr [1:4876] "" "" "" "" ...
##  $ prev_match_id              : chr [1:4876] NA "2022_M008" "2022_M024" "2022_M039" ...
##  $ prev_match_date            : Date[1:4876], format: NA "2022-11-22" ...
##  $ recovery_days              : int [1:4876] NA 4 4 3 6 9 NA 4 4 3 ...
##  $ prev_nominal_minutes_played: int [1:4876] NA 19 90 59 72 78 NA 31 27 79 ...
##  $ prev_total_distance_m      : num [1:4876] NA 3237 9991 6790 7821 ...
##  $ prev_zone1_distance_m      : num [1:4876] NA 1220 3888 2330 2693 ...
##  $ prev_zone2_distance_m      : num [1:4876] NA 1302 4326 3036 3625 ...
##  $ prev_zone3_distance_m      : num [1:4876] NA 476 974 847 881 ...
##  $ prev_zone4_distance_m      : num [1:4876] NA 171 552 452 436 ...
##  $ prev_zone5_distance_m      : num [1:4876] NA 69 251 126 186 ...
##  $ prev_top_speed_kmh         : num [1:4876] NA 31.9 29.6 30.7 29.5 31.6 NA 31.6 30 32.3 ...
##  $ normalized_total_distance_m: num [1:4876] 170 111 115 109 106 ...
##  $ normalized_zone1_distance_m: num [1:4876] 64.2 43.2 39.5 37.4 40.6 ...
##  $ normalized_zone2_distance_m: num [1:4876] 68.5 48.1 51.5 50.3 43.8 ...
##  $ normalized_zone3_distance_m: num [1:4876] 25 10.8 14.4 12.2 13 ...
##  $ normalized_zone4_distance_m: num [1:4876] 8.99 6.13 7.65 6.06 6.56 ...
##  $ normalized_zone5_distance_m: num [1:4876] 3.63 2.79 2.14 2.58 2.09 ...
```

2. Data Visualization

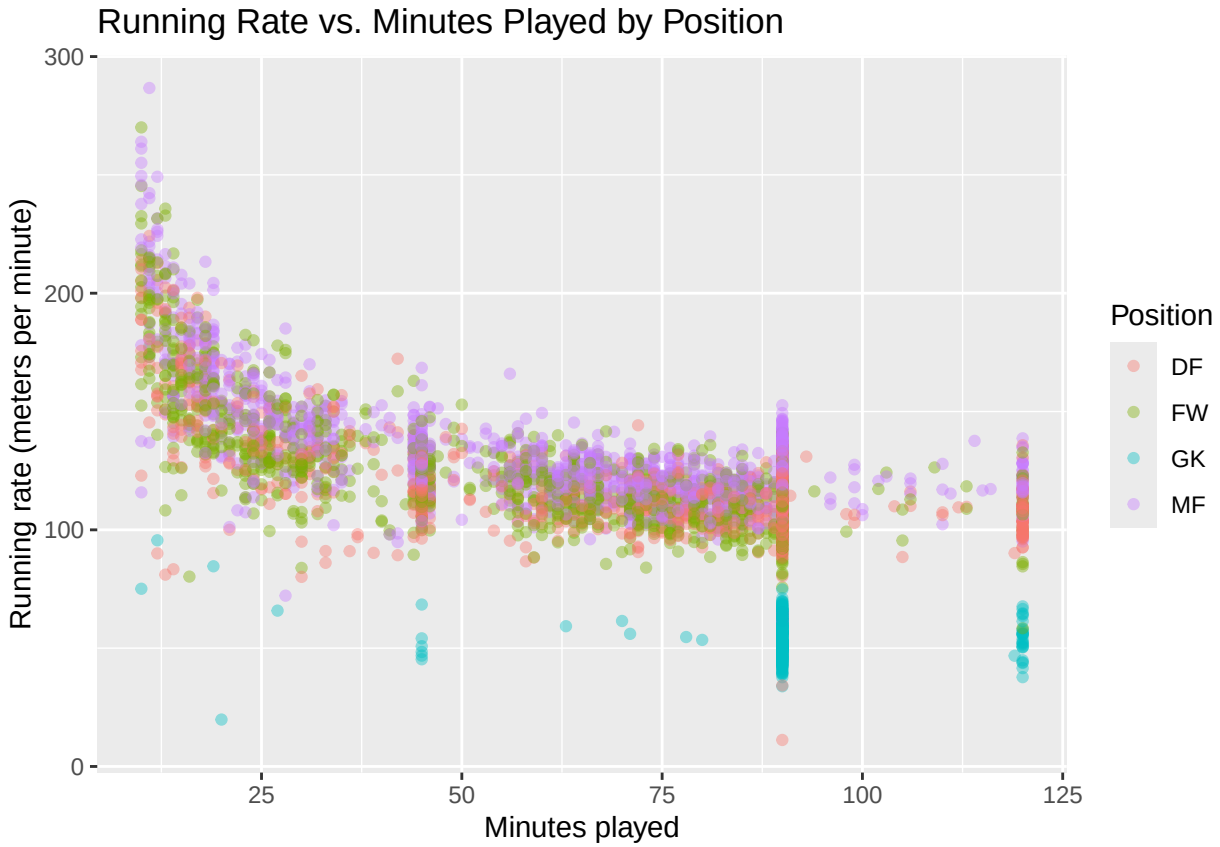
```r
# Running Rate Distribution by Position
df %>%
  ggplot(aes(
    x = normalized_total_distance_m,
    color = position,
    fill = position
  )) +
  geom_density(alpha = 0.15) +
  labs(
    title = "Running Rate Distribution by Position",
    x = "Running rate (meters per minute)",
    y = "Density",
    color = "Position",
    fill = "Position"
  )
```




```
pairs.panels(df[, c(
  "prev_nominal_minutes_played",
  "prev_total_distance_m",
  "recovery_days",
  "nominal_minutes_played",
  "normalized_total_distance_m"
)], gap = 0, pch = ".")
```

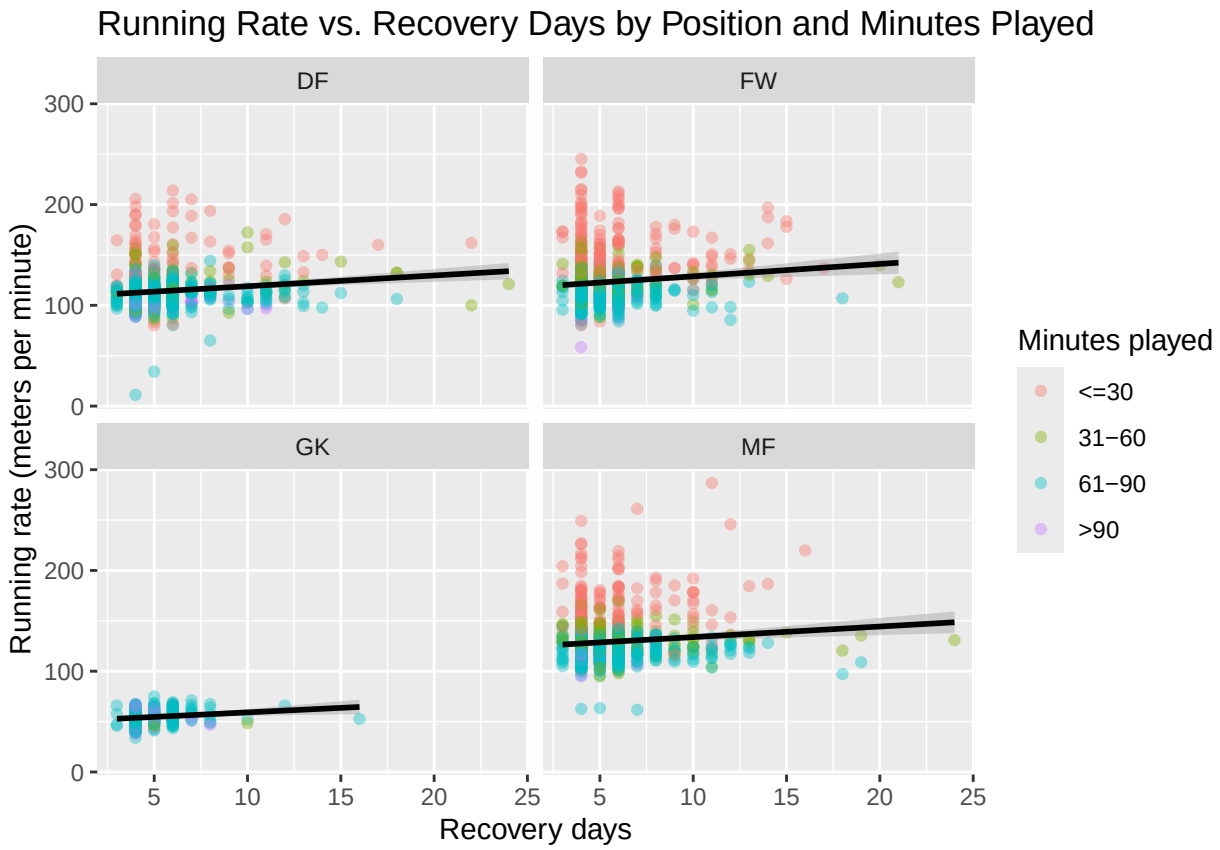



```r
# Running Rate vs. Minutes Played by Position
df %>%
  ggplot(aes(
    x = nominal_minutes_played,
    y = normalized_total_distance_m,
    color = position
  )) +
  geom_point(alpha = 0.4) +
  labs(
    title = "Running Rate vs. Minutes Played by Position",
    x = "Minutes played",
    y = "Running rate (meters per minute)",
    color = "Position"
  )
```




```
# Running Rate vs. Recovery Days by Position and Minutes Played
df %>%
  filter(!is.na(recovery_days)) %>%
  # filter(recovery_days < 15) %>%
  mutate(
    minutes_bin = cut(
      nominal_minutes_played,
      breaks = c(0, 30, 60, 90, Inf),
      labels = c(" 30", "31-60", "61-90", ">90")
    )
  ) %>%
  ggplot(aes(
    x = recovery_days,
    y = normalized_total_distance_m,
    color = minutes_bin
  )) +
  geom_point(alpha = 0.4) +
  geom_smooth(aes(group = 1), method = "lm", color = "black") +
  facet_wrap(~position) +
  labs(
    title = "Running Rate vs. Recovery Days by Position and Minutes Played",
    x = "Recovery days",
    y = "Running rate (meters per minute)",
    color = "Minutes played"
  )
```

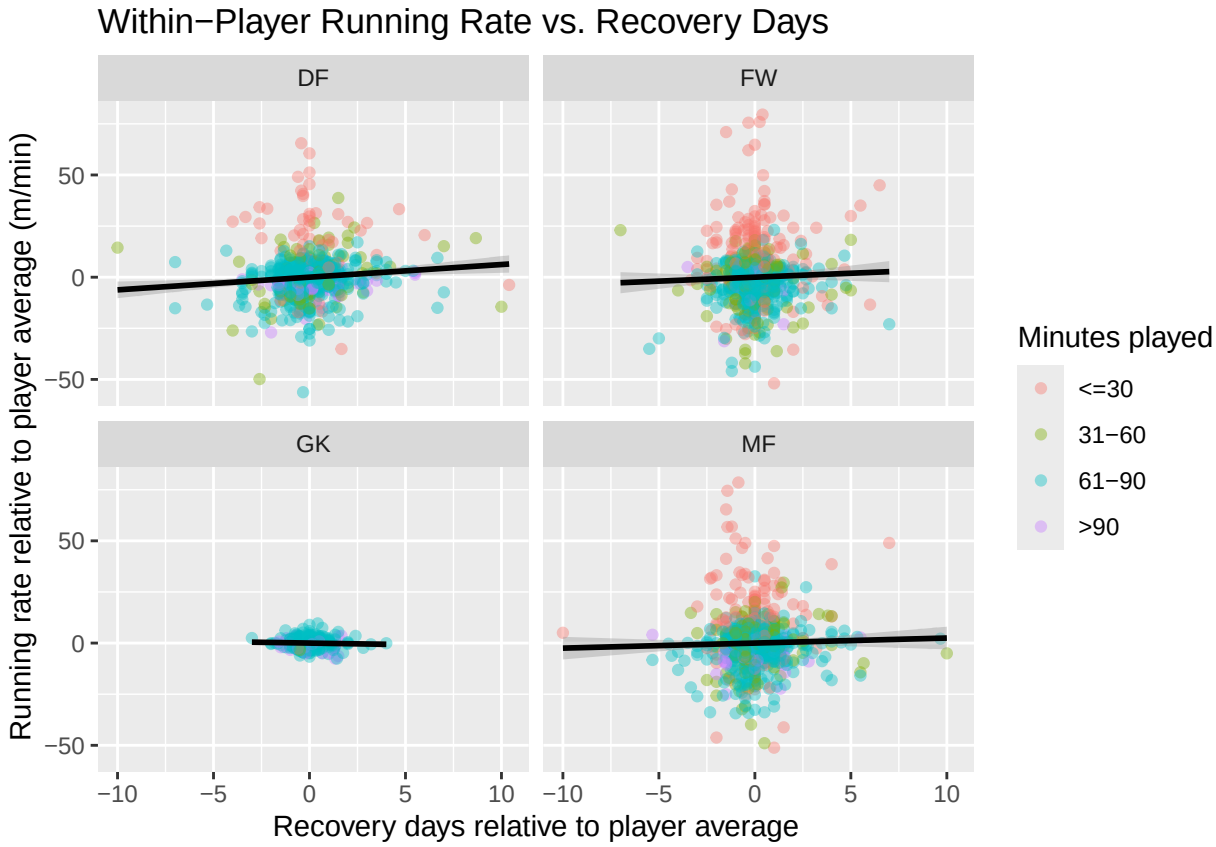
```
## `geom_smooth()` using formula = 'y ~ x'
```




```r
# Within-Player Running Rate vs. Recovery Days
df %>%
  filter(!is.na(recovery_days)) %>%
  group_by(player_tournament_id) %>%
  filter(n() >= 2) %>%
  mutate(
    recovery_dev = recovery_days - mean(recovery_days),
    running_rate_dev = normalized_total_distance_m - mean(normalized_total_distance_m)
  ) %>%
  ungroup() %>%
  mutate(
    minutes_bin = cut(
      nominal_minutes_played,
      breaks = c(0, 30, 60, 90, Inf),
      labels = c(" 30", "31-60", "61-90", ">90")
    )
  ) %>%
  ggplot(aes(
    x = recovery_dev,
    y = running_rate_dev,
    color = minutes_bin
  )) +
  geom_point(alpha = 0.4) +
  geom_smooth(aes(group = 1), method = "lm", color = "black") +
  facet_wrap(~position) +
  labs(
    title = "Within-Player Running Rate vs. Recovery Days",
```

```
    x = "Recovery days relative to player average",
    y = "Running rate relative to player average (m/min)",
    color = "Minutes played"
  )
```

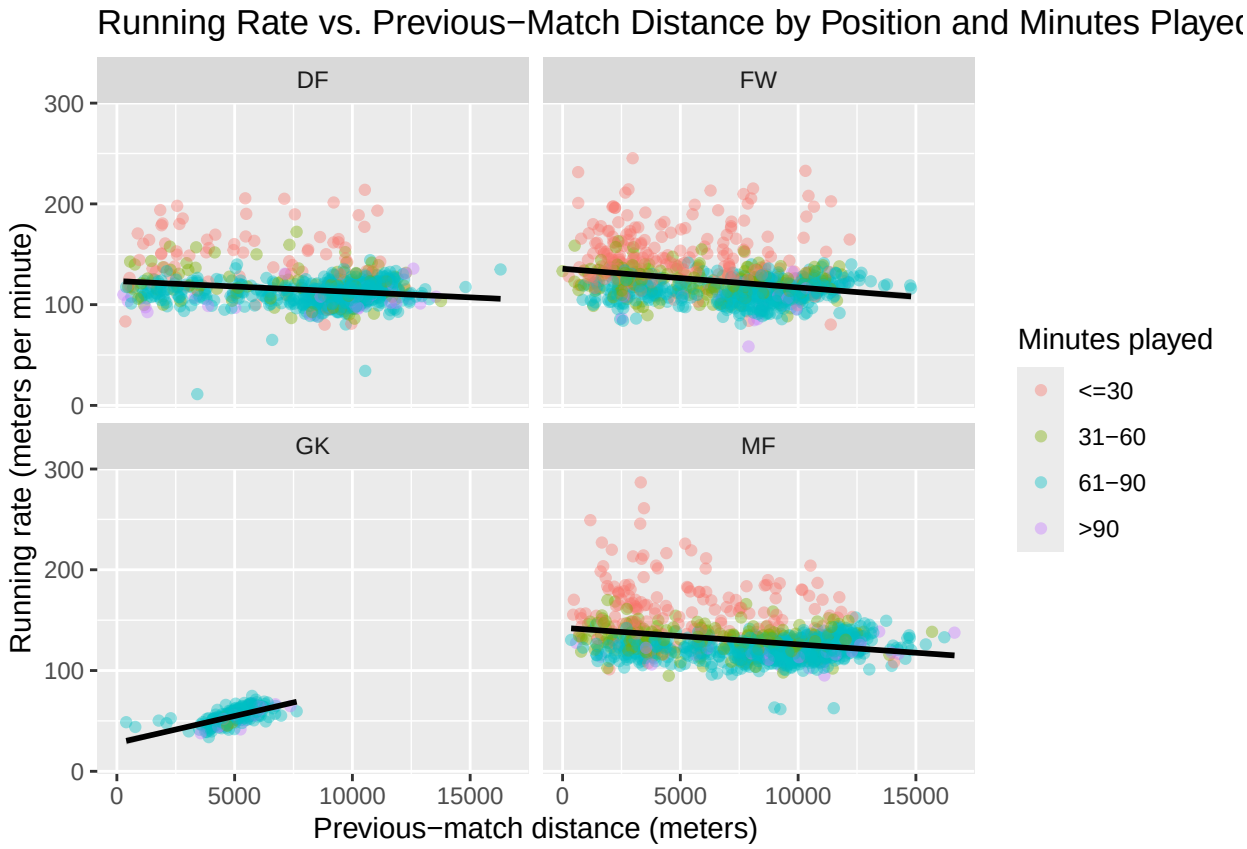
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# Running Rate vs. Previous-Match Distance by Position and Minutes Played
df %>%
  filter(!is.na(prev_total_distance_m)) %>%
  mutate(
    minutes_bin = cut(
      nominal_minutes_played,
      breaks = c(0, 30, 60, 90, Inf),
      labels = c(" 30", "31-60", "61-90", ">90")
    )
  ) %>%
  ggplot(aes(
    x = prev_total_distance_m,
    y = normalized_total_distance_m,
    color = minutes_bin
  )) +
  geom_point(alpha = 0.4) +
  geom_smooth(aes(group = 1), method = "lm", color = "black") +
  facet_wrap(~position) +
  labs(
    title = "Running Rate vs. Previous-Match Distance by Position and Minutes Played",
```

```
    x = "Previous-match distance (meters)",
    y = "Running rate (meters per minute)",
    color = "Minutes played"
  )
```

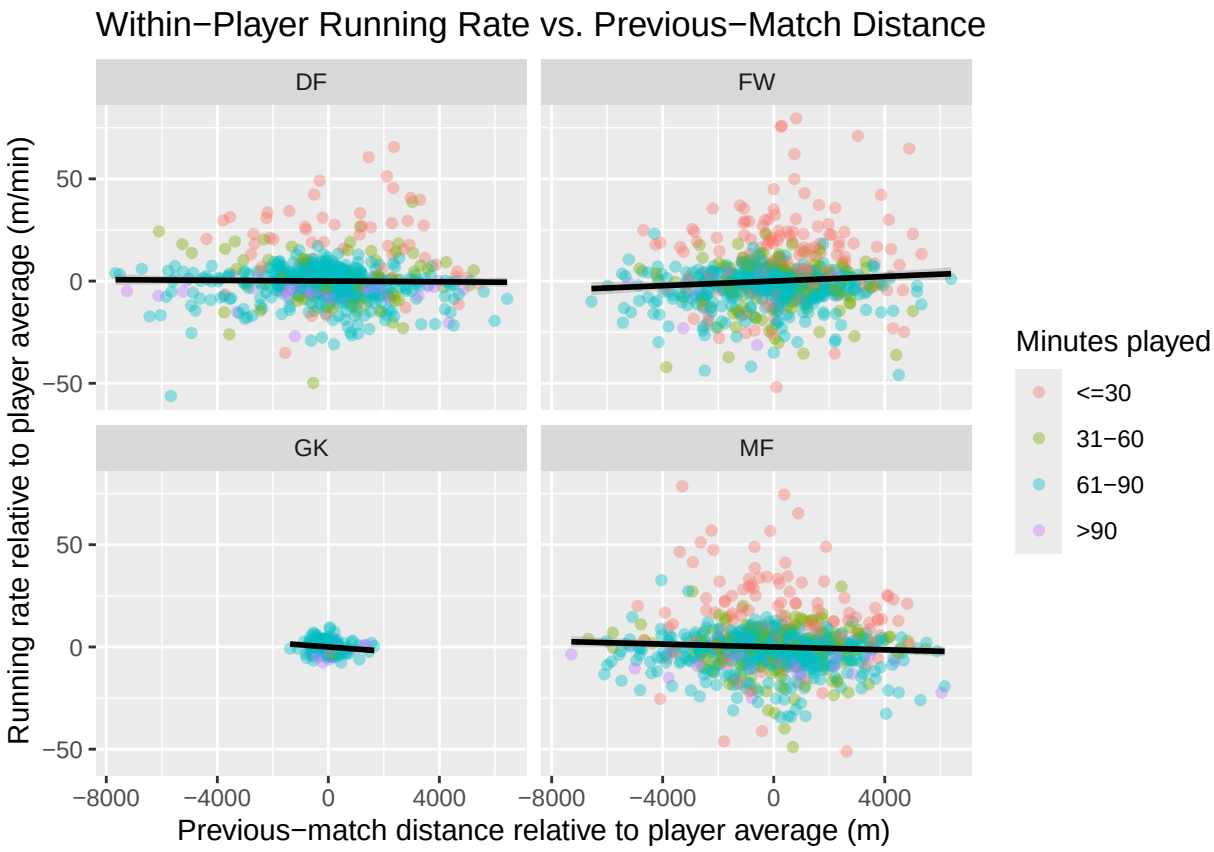
```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# Within-Player Running Rate vs. Previous-Match Distance
df %>%
  filter(!is.na(prev_total_distance_m)) %>%
  group_by(player_tournament_id) %>%
  filter(n() >= 2) %>%
  mutate(
    prev_distance_dev = prev_total_distance_m - mean(prev_total_distance_m),
    running_rate_dev = normalized_total_distance_m - mean(normalized_total_distance_m)
  ) %>%
  ungroup() %>%
  mutate(
    minutes_bin = cut(
      nominal_minutes_played,
      breaks = c(0, 30, 60, 90, Inf),
      labels = c(" 30", "31-60", "61-90", ">90")
    )
  ) %>%
  ggplot(aes(
    x = prev_distance_dev,
    y = running_rate_dev,
```

```
    color = minutes_bin
  )) +
  geom_point(alpha = 0.4) +
  geom_smooth(aes(group = 1), method = "lm", color = "black") +
  facet_wrap(~position) +
  labs(
    title = "Within-Player Running Rate vs. Previous-Match Distance",
    x = "Previous-match distance relative to player average (m)",
    y = "Running rate relative to player average (m/min)",
    color = "Minutes played"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



3. Modeling

```
# Ignore rows corresponding to first game at the tournament for a player (missing previous game data)
df_filtered <- df %>% filter(!is.na(recovery_days))
```

```
# normalized_total_distance_m ~ 1 + recovery_days + position
fit1 <- lm(normalized_total_distance_m ~ recovery_days + position, data = df_filtered)
summary(fit1)
```

```
##
## Call:
## lm(formula = normalized_total_distance_m ~ recovery_days + position,
##     data = df_filtered)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -101.268  -10.542   -2.960    5.442  151.551
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   108.0601     1.0266 105.255  < 2e-16 ***
## recovery_days   1.1017     0.1572   7.008  2.9e-12 ***
## positionFW      9.0776     0.8541  10.628  < 2e-16 ***
## positionGK    -58.9436     1.3516 -43.609  < 2e-16 ***
## positionMF     14.9882     0.8094  18.518  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 19.06 on 3363 degrees of freedom
## Multiple R-squared:  0.4881, Adjusted R-squared:  0.4875
## F-statistic: 801.8 on 4 and 3363 DF,  p-value: < 2.2e-16
```

```
# Add ns(nominal_minutes_played, df=4)
fit2 <- lm(normalized_total_distance_m ~ recovery_days + position + ns(nominal_minutes_played, df=4), da
summary(fit2)
```

```
##
## Call:
## lm(formula = normalized_total_distance_m ~ recovery_days + position +
##     ns(nominal_minutes_played, df = 4), data = df_filtered)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -99.168  -6.940   0.204   7.101 107.726
##
## Coefficients:
##                                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         165.5714     1.2355 134.015  < 2e-16 ***
## recovery_days                         0.2757     0.1094   2.519   0.0118 *
## positionFW                            2.5270     0.6265   4.034 5.61e-05 ***
## positionGK                          -55.5940     0.9380 -59.272  < 2e-16 ***
## positionMF                           12.0994     0.5749  21.047  < 2e-16 ***
## ns(nominal_minutes_played, df = 4)1  -55.4714     1.1764 -47.152  < 2e-16 ***
## ns(nominal_minutes_played, df = 4)2  -28.0561     1.0859 -25.838  < 2e-16 ***
## ns(nominal_minutes_played, df = 4)3 -119.0524     2.8510 -41.759  < 2e-16 ***
## ns(nominal_minutes_played, df = 4)4  -28.2607     1.1785 -23.980  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.11 on 3359 degrees of freedom
## Multiple R-squared:  0.7582, Adjusted R-squared:  0.7576
## F-statistic:  1317 on 8 and 3359 DF,  p-value: < 2.2e-16
```

```
# Add prev_total_distance_m + recovery_days * prev_total_distance_m
fit3 <- lm(normalized_total_distance_m ~ recovery_days + position + prev_total_distance_m + recovery_day
summary(fit3)
```

```
##
```

```
## Call:
## lm(formula = normalized_total_distance_m ~ recovery_days + position +
##     prev_total_distance_m + recovery_days * prev_total_distance_m +
##     ns(nominal_minutes_played, df = 4), data = df_filtered)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -97.748  -6.982   0.228   7.089 107.075
##
## Coefficients:
##                                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        1.626e+02  1.691e+00  96.133  < 2e-16 ***
## recovery_days                      6.503e-01  2.196e-01   2.961  0.00308 **
## positionFW                         2.693e+00  6.299e-01   4.276 1.96e-05 ***
## positionGK                        -5.469e+01  1.006e+00 -54.385  < 2e-16 ***
## positionMF                         1.206e+01  5.745e-01  20.999  < 2e-16 ***
## prev_total_distance_m              4.880e-04  1.860e-04   2.623  0.00875 **
## ns(nominal_minutes_played, df = 4)1 -5.603e+01  1.204e+00 -46.550  < 2e-16 ***
## ns(nominal_minutes_played, df = 4)2 -2.888e+01  1.140e+00 -25.330  < 2e-16 ***
## ns(nominal_minutes_played, df = 4)3 -1.204e+02  2.899e+00 -41.523  < 2e-16 ***
## ns(nominal_minutes_played, df = 4)4 -2.887e+01  1.213e+00 -23.804  < 2e-16 ***
## recovery_days:prev_total_distance_m -5.600e-05  3.073e-05  -1.822  0.06848 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.09 on 3357 degrees of freedom
## Multiple R-squared:  0.7588, Adjusted R-squared:  0.7581
## F-statistic:  1056 on 10 and 3357 DF,  p-value: < 2.2e-16
```

```
# Add random_effect(team)
fit4 <- lmer(normalized_total_distance_m ~ recovery_days + position + prev_total_distance_m + recovery_
```

```
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```

```
summary(fit4)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula:
## normalized_total_distance_m ~ recovery_days + position + prev_total_distance_m +
##     recovery_days * prev_total_distance_m + ns(nominal_minutes_played,
##     df = 4) + (1 | team_tournament_id)
##    Data: df_filtered
##
## REML criterion at convergence: 26790
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -7.3752 -0.5120  0.0099  0.5122  8.1016
##
## Random effects:
##  Groups             Name        Variance Std.Dev.
##  team_tournament_id (Intercept)  11.89    3.449
##  Residual                       160.70   12.677
## Number of obs: 3368, groups:  team_tournament_id, 80
```

```
##
## Fixed effects:
##                                      Estimate Std. Error t value
## (Intercept)                          1.624e+02  1.696e+00  95.770
## recovery_days                        7.143e-01  2.141e-01   3.337
## positionFW                           2.548e+00  6.159e-01   4.137
## positionGK                          -5.490e+01  9.755e-01 -56.280
## positionMF                           1.210e+01  5.605e-01  21.594
## prev_total_distance_m                4.490e-04  1.827e-04   2.457
## ns(nominal_minutes_played, df = 4)1 -5.589e+01  1.169e+00 -47.820
## ns(nominal_minutes_played, df = 4)2 -2.878e+01  1.114e+00 -25.838
## ns(nominal_minutes_played, df = 4)3 -1.192e+02  2.850e+00 -41.815
## ns(nominal_minutes_played, df = 4)4 -2.805e+01  1.205e+00 -23.268
## recovery_days:prev_total_distance_m -5.546e-05  3.028e-05  -1.831
##
## Correlation of Fixed Effects:
##             (Intr) rcvry_ pstnFW pstnGK pstnMF prv___ n(__,d=4)1 n(__,d=4)2
## recovry_dys -0.733
## positionFW  -0.287  0.034
## positionGK  -0.197  0.074  0.245
## positionMF  -0.198  0.000  0.503  0.233
## prv_ttl_ds_ -0.651  0.789  0.054  0.201 -0.023
## ns(__,d=4)1 -0.238  0.037 -0.043 -0.066 -0.041 -0.087
## ns(__,d=4)2 -0.248  0.013  0.267 -0.212  0.213 -0.156 -0.128
## ns(__,d=4)3 -0.445 -0.007  0.112 -0.113  0.064 -0.106  0.223      0.585
## ns(__,d=4)4 -0.073  0.042  0.114 -0.151  0.102 -0.085  0.314      0.154
## rcvry_d:___  0.588 -0.849 -0.003 -0.048  0.019 -0.899 -0.008      0.025
##             n(__,d=4)3 n(__,d=4)4
## recovry_dys
## positionFW
## positionGK
## positionMF
## prv_ttl_ds_
## ns(__,d=4)1
## ns(__,d=4)2
## ns(__,d=4)3
## ns(__,d=4)4  0.190
## rcvry_d:___  0.028     -0.020
## fit warnings:
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```

```
# Add random_effect(player)
fit5 <- lmer(normalized_total_distance_m ~ recovery_days + position + prev_total_distance_m + recovery_
```

```
## Warning: Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```

```
summary(fit5)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula:
## normalized_total_distance_m ~ recovery_days + position + prev_total_distance_m +
##     recovery_days * prev_total_distance_m + ns(nominal_minutes_played,
##     df = 4) + (1 | team_tournament_id) + (1 | player_tournament_id)
```

```
##    Data: df_filtered
##
## REML criterion at convergence: 26330
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -5.8561 -0.4182 -0.0006  0.4141  6.9344
##
## Random effects:
##  Groups               Name        Variance Std.Dev.
##  player_tournament_id (Intercept) 79.43    8.912
##  team_tournament_id   (Intercept) 10.36    3.218
##  Residual                         91.88    9.585
## Number of obs: 3368, groups:
## player_tournament_id, 1341; team_tournament_id, 80
##
## Fixed effects:
##                                      Estimate Std. Error t value
## (Intercept)                          1.632e+02  1.696e+00  96.258
## recovery_days                        5.098e-01  2.024e-01   2.518
## positionFW                           3.384e+00  8.311e-01   4.072
## positionGK                          -5.744e+01  1.329e+00 -43.213
## positionMF                           1.251e+01  7.634e-01  16.389
## prev_total_distance_m               -2.792e-05  1.694e-04  -0.165
## ns(nominal_minutes_played, df = 4)1 -5.383e+01  1.071e+00 -50.257
## ns(nominal_minutes_played, df = 4)2 -2.526e+01  1.025e+00 -24.632
## ns(nominal_minutes_played, df = 4)3 -1.130e+02  2.568e+00 -44.007
## ns(nominal_minutes_played, df = 4)4 -2.592e+01  1.024e+00 -25.322
## recovery_days:prev_total_distance_m -2.743e-05  2.799e-05  -0.980
##
## Correlation of Fixed Effects:
##             (Intr) rcvry_ pstnFW pstnGK pstnMF prv___ n(__,d=4)1 n(__,d=4)2
## recovry_dys -0.707
## positionFW  -0.338  0.026
## positionGK  -0.200  0.049  0.261
## positionMF  -0.268  0.000  0.492  0.268
## prv_ttl_ds_ -0.669  0.801  0.072  0.124  0.009
## ns(__,d=4)1 -0.269  0.038  0.008 -0.021 -0.016 -0.033
## ns(__,d=4)2 -0.303  0.040  0.222 -0.116  0.157 -0.047 -0.125
## ns(__,d=4)3 -0.450  0.006  0.099 -0.060  0.047 -0.050  0.250      0.585
## ns(__,d=4)4 -0.142  0.060  0.118 -0.073  0.087  0.001  0.314      0.228
## rcvry_d:___  0.578 -0.861 -0.009 -0.028  0.012 -0.899 -0.016     -0.010
##             n(__,d=4)3 n(__,d=4)4
## recovry_dys
## positionFW
## positionGK
## positionMF
## prv_ttl_ds_
## ns(__,d=4)1
## ns(__,d=4)2
## ns(__,d=4)3
## ns(__,d=4)4  0.192
## rcvry_d:___  0.011     -0.051
## fit warnings:
```

```
## Some predictor variables are on very different scales: consider rescaling.
## You may also use (g)lmerControl(autoscale = TRUE) to improve numerical stability.
```